

alertsreportshotspotsusers+

google > google > alerts

Open MongoDB shell

Documents16AggregationsSchemaIndexes1Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATABULKEXPORT DATAEXPORT CODE

251-16 of 16

```
_id: ObjectId('6a4f17f0c7809143ac86ab2e')
reportId: ObjectId('6a4f17f0c7809143ac86ab29')
pollutionType: "Clean Environment"
location: "Lat: 13.0421, Lng: 80.1746"
latitude: 13.0421
longitude: 80.1746
AQI: 25
predictedAQI: 62
confidence: 100
priority: "High"
reason: "The image shows a pristine forest environment with dense trees and lus..."
suggestedAction: "Dispatch General Inspection Team"
status: "Pending"
assignedTeam: null
createdAt: 2026-07-09T03:39:28.850+00:00
__v: 0
```

```
_id: ObjectId('6a4f1a0ec7809143ac86ab8c')
reportId: ObjectId('6a4f1a0ec7809143ac86ab87')
pollutionType: "Garbage Burning"
location: "Lat: 11.1441, Lng: 77.3250"
latitude: 11.144082
longitude: 77.324973
AQI: 25
predictedAQI: 84
confidence: 100
priority: "Critical"
reason: "The image clearly shows an open pile of mixed waste (including plastic..."
suggestedAction: "Deploy Fire & Cleanup Team"
status: "Pending"
assignedTeam: null
createdAt: 2026-07-09T03:48:30.769+00:00
__v: 0
```

Options ▶

25 1 - 16 of 16

▼ Show 20 more fields

```
_id: ObjectId('64ae7142b8bf25a830c96c6d')
citizenId: null
reportedBy: ObjectId('64ae7133b8bf25a830c96c64')
title: ""
image: "/api/reports/64ae7142b8bf25a830c96c6d/image"
```

alertsreportshotspotsusers+

google > google > hotspots

>_ Open MongoDB shell

Documents 4AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 4 of 4

```
_id: ObjectId('6a4f1a1cc7809143ac86abb0')
centerLat: 11.143815887213758
centerLng: 77.32498106511724
radius: 300
reportCount: 4
averageAQI: 25
averageConfidence: 99
highestSeverity: "Critical"
risk: "Critical"
dominantPollution: "Clean Environment"
recommendedAction: "Dispatch Emergency Response Team"
assignedTeam: null
assignedOfficer: null
assignedOfficerName: ""
location: "Lat: 11.1437, Lng: 77.3250"
sourceReportIds: Array (4)
sourceReportData: Array (4)
municipalStatus: "pending"
status: "Active"
createdAt: 2026-07-09T03:48:44.193+00:00
__v: 0
```

```
_id: ObjectId('6a4f1a1cc7809143ac86abb1')
centerLat: 9.912533333333334
centerLng: 78.11433333333333
radius: 357
reportCount: 3
averageAQI: 25
averageConfidence: 99
highestSeverity: "Critical"
risk: "Critical"
dominantPollution: "Factory Emission"
```

alertsreportshotspotsusers+

google > google > users

> Open MongoDB shell

Documents 10AggregationsSchemaIndexes 2Validation

Type a query: { field: 'value' } or [Generate query](#)

ExplainResetFindOptions

ADD DATABULKEXPORT DATAEXPORT CODE

251 - 10 of 10

```
_id: ObjectId('6a4e76b906a8ed38de8eb71c')
name: "24ALR044005"
email: "24alr044005@madurai4nation.com"
password: "$2a$10$A3MsMvq52yuw2PvK/PDGw.08KdRxdU0so/XIzB6exlB0Ne8SwVKBG"
role: "citizen"
status: "active"
department: ""
createdAt: 2026-07-08T16:11:37.476+00:00
__v: 0
```

```
_id: ObjectId('6a4e76c106a8ed38de8eb72d')
name: "24ALR044006"
email: "24alr044006@madurai4nation.com"
password: "$2a$10$twUAG3s/hoWDN0hlrzo64u7iWUjvgirLXtER5/6d2mZ4bWSlxqt.q"
role: "citizen"
status: "active"
department: ""
createdAt: 2026-07-08T16:11:45.129+00:00
__v: 0
```

```
_id: ObjectId('6a4f167065b7a373899d2d24')
name: "Citizen Demo"
email: "citizen@madurai4nation.com"
password: "$2a$10$Abq/wgb2KXHvt0KRtnk8T0LiLiCsR.U9F.CPfV05PNx04qzG9dfq"
role: "citizen"
status: "active"
department: ""
createdAt: 2026-07-09T03:33:04.482+00:00
__v: 0
```

AI-Powered Municipal Air Intelligence

Protect Your City. One Report. One Click.

AI Detects Pollution Before It Becomes Dangerous. Citizens report, AI analyzes, and municipalities act — all in real-time.

Report Pollution →

View Dashboard ↗

RADAR: HIGH RESOLUTION

GRID_OK: 99.8%



PM2.5 Index

82 $\mu\text{g}/\text{m}^3$

Response Rate

4.2 min

REPORTS TODAY

47

+12.5%

vs previous month



ACTIVE HOTSPOTS

23

-8.3%

vs previous month



AI ACCURACY

96

+2.1%

vs previous month



AVG RESPONSE TIME

4

+18%

vs previous month



Upload a photo. Gemini 2.5 Flash inspects it. Data feeds the dashboard and map.

 Live Camera

Lat: 11.1455, Lng: 77.3254

Dense vehicle congestion with visible smog layer.

Run **Analyze with AI** to inspect and save to database.

OpenWeather data for your coordinates.

AQI data unavailable

🔗 GEMINI_2.5_FLASH

Awaiting AI Assessment

[Upload a pollution photo](#)

CITIZEN PORTAL

My Reports

Track the municipal response progress for your submitted pollution reports.

🔄 Refresh

4

TOTAL REPORTS

0

RESOLVED

0

IN PROGRESS



Vehicle Poll...

high

AQI 25

CONFIDENCE

95%

HEALTH RISK

High conc...

Report Submitted

09 Jul 2026, 09:43 am
Lat: 11.1455, Lng: 77.3254

Garbage Bu...

critical

AQI 25

CONFIDENCE

100%

HEALTH RISK

Inhalation ...

Report Submitted

09 Jul 2026, 09:18 am
Lat: 11.1441, Lng: 77.3250

Clean Envir...

low

AQI 25

CONFIDENCE

100%

HEALTH RISK

None

Report Submitted

09 Jul 2026, 09:09 am
Lat: 13.0421, Lng: 80.1746

Vehicle Poll...

high

AQI 25

CONFIDENCE

98%

HEALTH RISK

High risk o...

Report Submitted

09 Jul 2026, 09:03 am
Lat: 13.0421, Lng: 80.1746

CleanAir AI

An advanced, hyperlocal climate intelligence and pollution monitoring network. Empowering citizens and municipalities to safeguard clean air using AI-assisted imagery and spatial analytics.



Navigation

[Home Page](#)[Report Pollution](#)[Air Quality Dashboard](#)[Live Spatial Map](#)[About & FAQ](#)

Contact Response

premkalai2006@gmail.com[9894503899](tel:9894503899)[Madurai, Tamil Nadu, India](#)**Team:** Global Impact Challenge

Legal & Policy

[Privacy Policy](#)[Terms of Service](#)[Municipal Data SLA](#)[API Accessibility](#)

17 Reports
Loaded

Hotspots are automatically clustered from reports in the last 24 hours and shown as semi-transparent circles with

17 Reports
Loaded

Real-time OpenStreetMap view of citizen-reported pollution across Madurai – from garbage burning to vehicle emissions, analyzed by Gemini AI and plotted from MongoDB.

Spatial Filters

Filter pollution reports across Madurai municipality.

SEARCH LOCATION

Q Search location or category...

SEVERITY

All Critical High Medium

Low

POLLUTION CATEGORY

All Categories

MAP LAYERS

- ✓ **Heatmap Overlay**

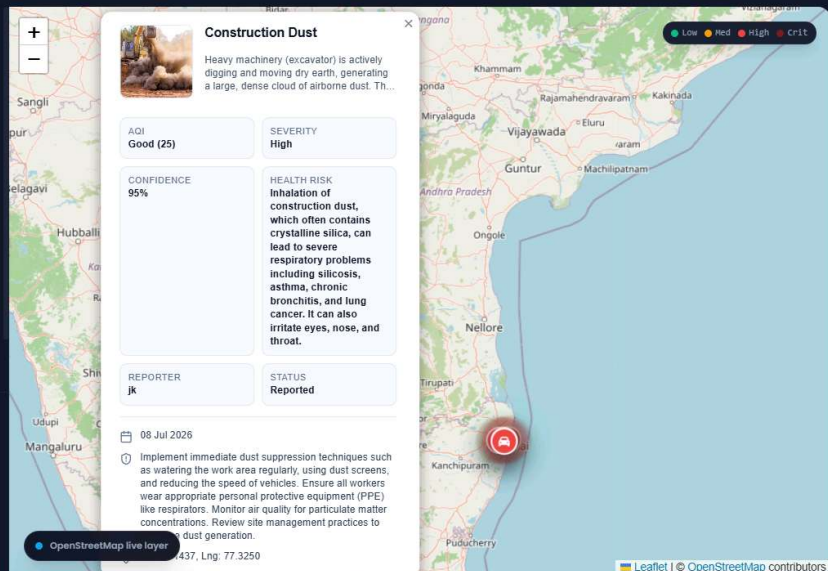
17 / 17

Active Hotspots

4

[Refresh from MongoDB](#)

Click any marker to view pollution details, AQI, and municipal recommendations.



Citizen Reports → Map

Every report submitted via photo upload is analyzed by Gemini AI, enriched with OpenWeather AQI data, saved to

Severity Color Coding

Green markers indicate low-risk pollution, yellow for medium severity, and red for high or critical incidents —

Hotspot Detection Engine

Hotspots are automatically clustered from reports in the last 24 hours and shown as semi-transparent circles with

AIR QUALITY INDEX

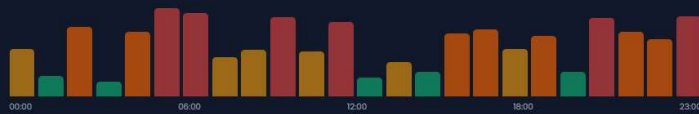
Live AQI & Prediction

Real-time air quality data and AI-powered 24-hour pollution forecast for Madurai.

CURRENT AQI

Moderate

25 AQI

PM2.5: 4.4 $\mu\text{g}/\text{m}^3$ 

24H PREDICTION

126 AQI

Trend: Increasing



Prediction Confidence

WEATHER CONDITIONS

TEMPERATURE

31.3°C

HUMIDITY

46%

WIND SPEED

4.5 m/s

NEARBY REPORTS

4

POLLUTANTS

PM2.5	4.4 $\mu\text{g}/\text{m}^3$
PM10	6.4 $\mu\text{g}/\text{m}^3$
NO ₂	3.1 ppb
SO ₂	2.0 ppb
CO	125.1 ppb
O ₃	26.4 ppb

AQI REFERENCE GUIDE

CleanAir AI

DashboardAll ReportsHotspot IntelligenceAnalyticsProfile

Gemini: 5/1500

Operations Center

Municipal pollution monitoring & response command

Thursday, 9 July 2026

09:45:39 am

Municipal Monitoring Active

AI Monitoring Enabled

OD

Good Morning, Officer

Officer

17

Total Cases

17

Pending

QUICK ACTIONS

Assign Team

Export Report

AI Summary

Emergency Broadcast

Refresh

8 Critical Alerts

LIVE

Lat: 11.1441, Lng: 77.3250

Avg Response: 4.2 min

17 unassigned

Last sync: 09:45:39 am

17 Pending

Review Critical Reports →

Performance Overview

All metrics updated in real-time

17

Total Reports

+12% ↑

Updated 09:45:39 am

17

Pending

+5% ↑

Updated 09:45:39 am

0

Verified

+8% ↑

Updated 09:45:39 am

0

Resolved

+15% ↑

Updated 09:45:39 am

6

High Priority

-2% ↓

Updated 09:45:39 am

8

Critical Priority

0% ↓

Updated 09:45:39 am

0

In Progress

+3% ↑

Updated 09:45:39 am

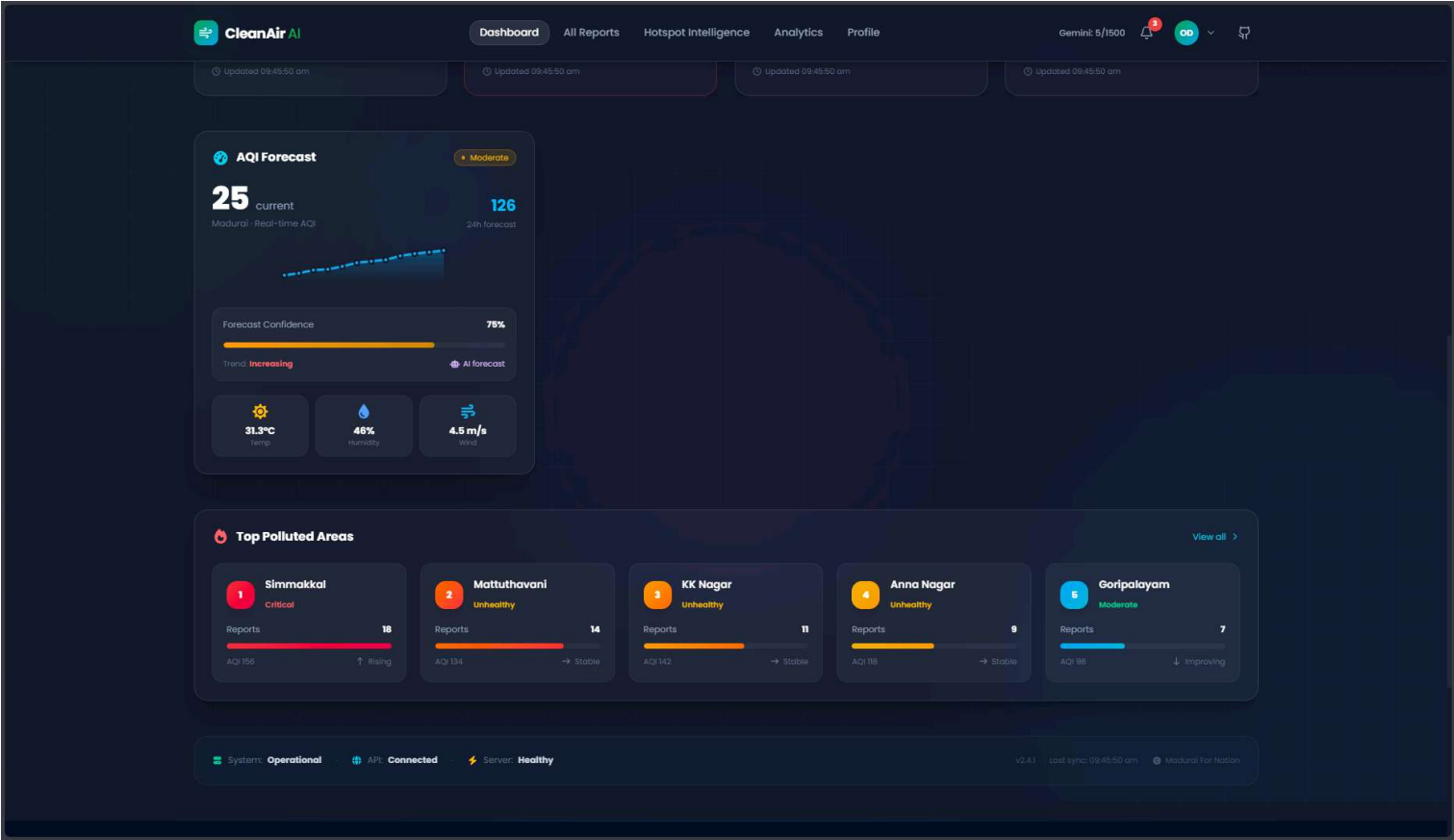
5

Today's Reports

+22% ↑

Updated 09:45:39 am

localhost:3000/#/officer/reports



CleanAir AI

DashboardAll ReportsHotspot IntelligenceAnalyticsProfile

Gemini: 5/1500

3

00

MUNICIPAL OPERATIONS

All Reports

Review, assign, and track every citizen pollution report across Madurai – filter by severity, monitor municipal response, and take action.

Refresh

Hotspot Map

TOTAL REPORTS

17

PENDING

14

IN PROGRESS

3

RESOLVED

0

HIGH PRIORITY

14

Q Search by category, location, reporter, or ID...

All Categories

Newest First

SEVERITY

All

Critical

High

Medium

Low

STATUS

All

Pending

In Progress

Resolved

Showing 17 of 17 reports

Critical

AQI 25

#7C657F

Vehicle Pollution

Dense vehicle congestion with visible smog layer.

Lat: 11.1455, Lng: 77.3254

Citizen Demo

09 Jul 2026, 09:43 am

Report Submitted

10%

View

Manage

Critical

AQI 25

#6A8B7

Garbage Burning

Industrial emissions from refinery area.

Lat: 11.1441, Lng: 77.3250

Citizen Demo

09 Jul 2026, 09:18 am

Report Submitted

10%

View

Manage

Low

AQI 25

#66AB29

Clean Environment

Clean coastal area with no visible pollution.

Lat: 13.0421, Lng: 80.1746

Citizen Demo

09 Jul 2026, 09:09 am

Report Submitted

10%

View

Manage

Report Details



CATEGORY

Vehicle Pollution

High Report Submitted

Location: 11.1455, lng: 77.3254

Reported by: Citizen Demo

Reported at: 09 Jul 2026, 09:43 am

Dense vehicle congestion with visible smog layer.

AIR QUALITY AT SITE

Live Air Quality

SOURCE: OPENWEATHER

AQI INDEX

25

GOOD

Temp

0.0°C

Humidity

0%

Wind

6 km/h

24h Trend
Last 12 readings



Very poor. Stay indoors with windows closed.

POLLUTANTS



Close

Manage Municipal Response

MUNICIPAL INTELLIGENCE

Hotspot Intelligence

Pollution clusters detected across Madurai from grouped citizen reports.

Refresh

All Reports



3 critical hotspots require immediate action

Assign response teams and advance municipal status promptly.

View First

ACTIVE HOTSPOTS

4



CRITICAL

3

HIGH RISK

1



AVG AQI

25



REPORTS CLUSTERED

12



NEEDS ACTION

4



Hotspots

4

Search zones...

All Critical High Medium

Low

All Pending Active

Resolved

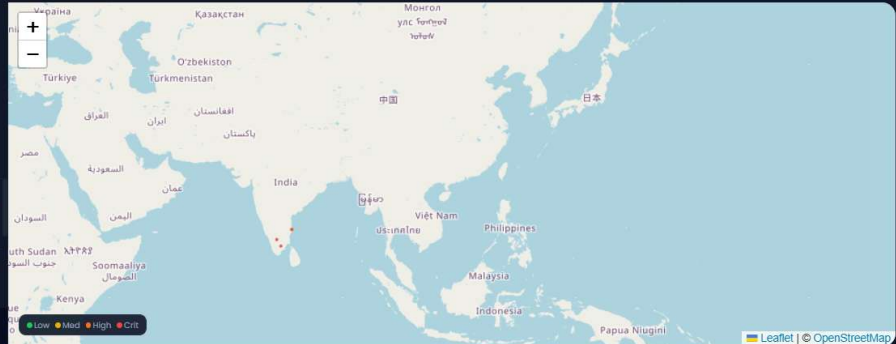
Lat: 11.1437, Lng: 77.321 Critical

5 reports - AQI 25

SIDCO Industrial Estate Critical

3 reports - AQI 25

Click a zone to inspect. Map zooms automatically.



Hotspot Zones

4 detected

Lat: 11.1437, Lng: 77.3250 Critical

5 reports - AQI 25

Clean Environment

Report Submitted

SIDCO Industrial Estate, Madurai Critical

3 reports - AQI 25

Factory Emission

Report Submitted

T Nagar, Chennai Critical

2 reports - AQI 25

Vehicle Pollution

Report Submitted

Lat: 13.0421, Lng: 80.1746 High

2 reports - AQI 25

Vehicle Pollution

Report Submitted

[Refresh](#)
[All Reports](#)

3 critical hotspots require immediate action
Assign response teams and advance municipal status promptly.

ACTIVE HOTSPOTS

CRITICAL
3

HIGH RISK

1



AVG AQI

25



REPORTS CLUSTERED

12

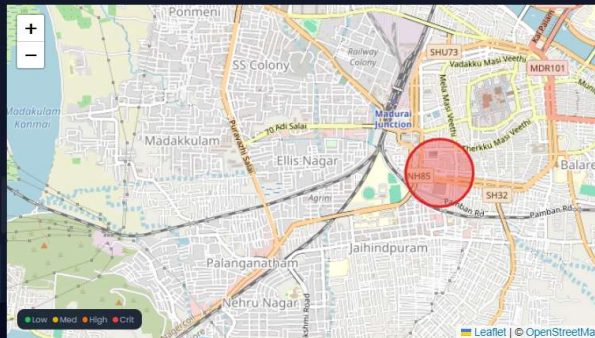
NEEDS ACTION
4

All Critical High Medium
Low
All Pending Active
Resolved

 **Lat: 11.1437, Lng: 77.321** Critical
5 reports AQI 25

 **SIDCO Industrial Estal** Critical

Click a zone to inspect - Map zooms automatically



09 Jul 2026, 09:46 am

Report Submitted 10%

REPORTS

AVG AQI

CONF.

RADIUS
OFF

Details

Lat: 11.1437, Lng: 77.3250

Report Submitted

SIDCO Industrial Estate, Madurai

Report Submitted

T Nagar, Chennai

Report Submitted

Lat: 13.0421, Lng: 80.1746

Report Submitted

SIDCO Industrial Estate, Madurai



Critical

Report Submitted

REPORTS

3

AVG AQI

25

CONFIDENCE

99%

RADIUS

357 m

DOMINANT POLLUTION

Factory Emission

RECOMMENDED ACTION

Escalate Industrial Pollution Inspection



Factory Emission



Factory Emission

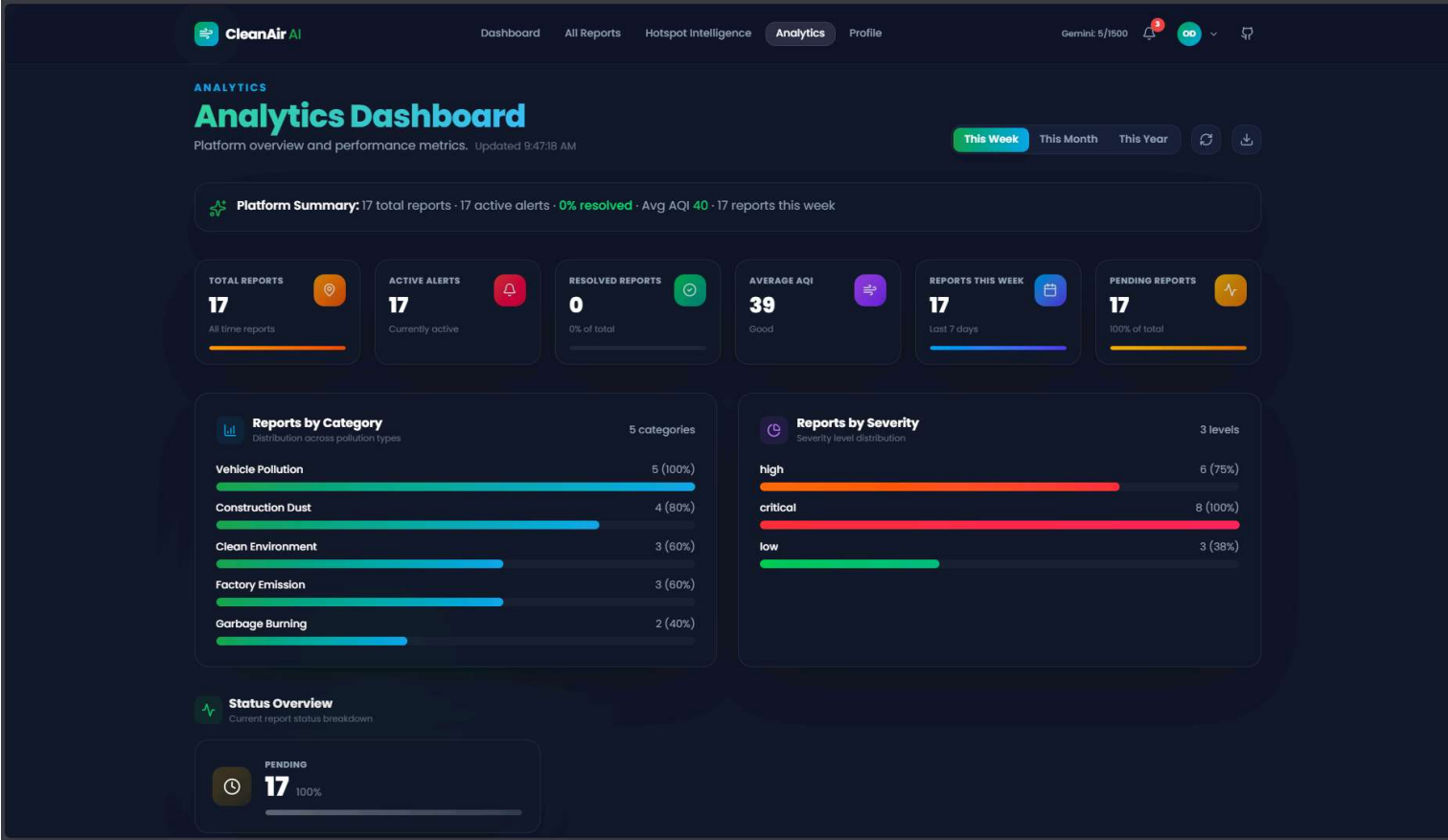


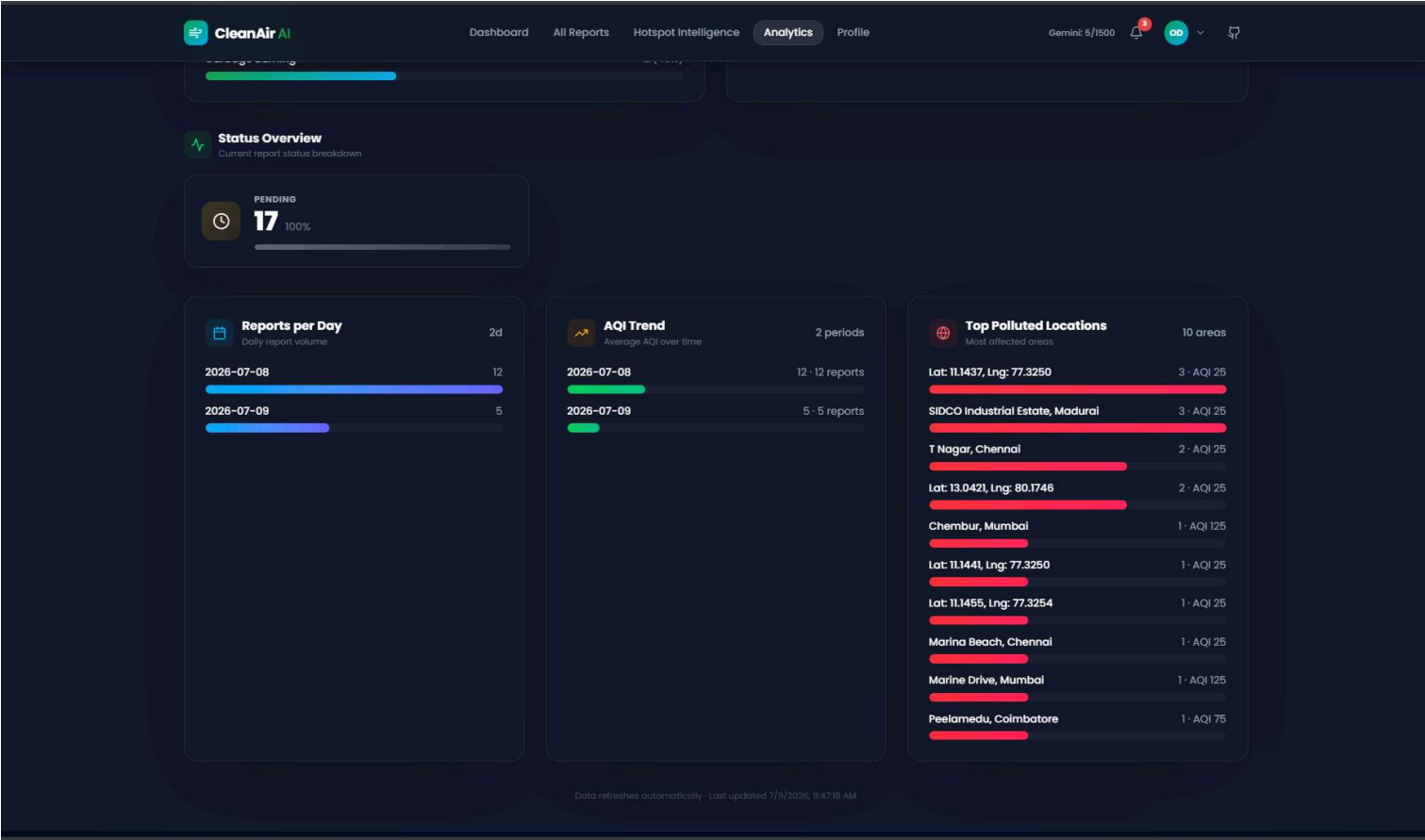
Construction Dust

9.9125, 78.1143

09 Jul 2026, 09:46 am

Close





CleanAir AI

DashboardAll ReportsHotspot IntelligenceAnalyticsProfile

Gemini: 5/1500

Account Overview

Manage your account details and preferences.

Edit Profile

OD

Officer Demo

Officer

Cleaner0 XP

Guardian - 100 XP away

Sign Out

Account Details

NAMEOfficer Demo

ROLEOfficer

DEPARTMENTEnvironmental Monitoring

EMAILofficer@madurai4nation.com

MEMBER SINCEJuly 9, 2026

REPORTS REVIEWED0

Click "Edit Profile" to update your details

REPORTS REVIEWED0

RESOLVED0

IN PROGRESS0

TOTAL REPORTS17

Achievements & Badges

Pollution HeroSubmitted 50+ pollution reports

Active ReporterReported in last 7 days

Green Citizen5 resolved cleanups

Environmental GuardianContributed to 25 reports

Impact Maker500+ people benefited

Contribution Activity

Last 30 days

LessMore